# Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

## Exercise 7.1 · Building a Right

The plane is equipped with an orthonormal coordinate system. $d$ go through $A(-1; 3)$ and admits the vector $\vec{n}$ = (2; 5) for normal vector.

1. Write the condition of belonging of a point $M(x; y)$ à $d$ using a scalar product. deduce a Cartesian equation from $d$.
2. Provide a non-zero policy vector of $d$, then the coordinates of a point of $d$ separate from $A$.
3. Check your point by substitution in the equation.

## Exercise 7.2 · Calculate a Project

In an orthonormal coordinate system, the point is given $P(4; 3)$ and the right $d$ d-equation $2x - y = 0$.
$H$ the orthogonal project of $P$ on $d$.

1. Determine the coordinates of $H$ by translating the two conditions: $H$ belongs to $d$ and $\overrightarrow{PH}$ is normal to $d$.
2. Check these two conditions separately with the coordinates obtained.
3. Calculate the distance from $P$ right $d$ in exact form.

### Exercise 7.3 · Recognize a circle

All points are considered $M(x; y)$ of the orthonormous plane whose coordinates check

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

1. Show that this set is a circle and specify its centre and radius.
2. Determine all its intersection points with the equation line $y = -2$.
3. Interpret geometrically the position of these two points on the circle.

### Exercise 7.4 · Ombudsperson

In an orthonormal coordinate system, we give $A(-1; 1)$ and $B(3; 3)$. An equation of the mediator is sought. of [ $AB$].

1. Determine this equation using the geometric properties of the mediator. choice of a point and a normal vector.
2. Verify that the midpoint of [$AB$] satisfies the equation obtained.

### Exercise 7.5 · Best Access

A straight road is represented by the right $d$ (d) equation $y = 2x$ in an orthonormal coordinate system of which unit of length is 1 km. A building is located at the point $P4; 3)$. We want to build a straight access connecting the building to a point of the road.

1. Determine the point of the road that makes this access as short as possible. exact access.
2. Justify that your choice is optimal across all points of the road, comparing distances using a right triangle. Only one observation on a drawing is not enough.

### Exercise 7.6 · Parameter and Set

For a real parameter $k$, we define $\mathcal{E}_k$ as all points $M(x; y)$ whose coordinates check

$$x^2 + y^2 - 2x + 4y + k = 0.$$

1. Determine the nature of the $\mathcal{E}_k$ based on the value of $k$. Organize the response in disjointed cases.
2. When this set is a circle, specify its centre and radius; in other cases, describe exactly the set obtained.

### Exercise 7.7 · A parabola as a locus

**Optional deepening.** In an orthonormal coordinate system, the point is fixed $F(0; 2)$. points $M(x; y)$ whose distance to $F$ is equal to the distance at the axis of the abscisses.

1. Express the square of each of the two distances according to $x$ and $y$.
2. Deduce an equation of the set sought. Verify the converse in order to justify this equation describes exactly all the points sought.

### Fiscal year 7.8 · Deferred review

In an orthonormal coordinate system, $\mathcal{C}$ is the circle of diameter [ $AB$], with $A(0; 0)$ and $B(4; 0)$.
the right $d$ (d) equation $x = 5$.

1. Determine whether $\mathcal{C}$ and $d$ have a common point, solving an equation.

2. Confirm the response by comparing the distance from the centre of the circle to the right and the radius of the circle circle.

**Personal work.** Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

## To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, Exercise 18 p. 279, questions a, b, c and d.** Read a normal non-zero vector for each equation, without solving in y.

**CQFD, Exercise 21 p. 279, questions a and b.** Write the condition of orthogonality to normal, then check that the given point belongs to the right.